

TSA - Data Science and Analytics

Documentation Portfolio

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Team 22391-3

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Introduction and Data Overview

The house prices all throughout the United States have been rising over the past years, which has a huge societal issue which affects people across the nation. The National Low Income Housing Coalition (NLIHC) describes that there is a shortage of over seven million affordable homes, specifically 7.3 million, for families with extremely low incomes. This leads to many people having issues finding housing for necessities. This extends beyond the problem of affordable housing and leads to homelessness. According to the article, “U.S. Homelessness Jumps to Another Record High, Amid Affordable Housing Shortage and Rising Costs,” Kline reports that more than 770,000 people in the U.S. are currently homeless. This issue is about rising home prices, but also about the challenges it creates for many people across the country.

The U.S. Department of Housing and Urban Development (HUD) states that the threshold for money spent on housing is 30%. The shortage of affordable housing has been a burden on many families because they spend more than the threshold. This leads to families not having much money for other necessities like healthcare, for example. Many families must constantly keep relocating due to this crisis.

This analysis can build a deeper understanding about housing rates and their correlation with income. By analyzing these rates, a greater understanding can be developed about lack of affordable housing across different regions in the United States.

Data Dictionary

Field Name	Data Type	Description	Field Example
Size Rank	Integer	Ranking of the size of a region based on the population	29
Region Name - State	String	Name of the state where housing data was collected from	Pennsylvania
Region Name - County	String	Name of the county where housing data was collected from	Los Angeles County
Region Type	String	Specific type of a region	County
Year	Integer	Year in which data was collected	2023
State Name	String	Full name of the state where the county is in	North Carolina
State	String	State abbreviation of where the county is in	NC
Metro	String	Metropolitan area in which the region is in	New York-Newark-Jersey City, NY-NJ-PA
Median Household Income	Float	Median value of the total income of a household in a specific region	\$130,000.00

Purpose

Problem Statement: There is an increasing number of people with the problem of a lack of affordable housing across the United States due to rising housing prices. This is important because it affects a huge portion of Americans.

Analysis: The analysis from this research is to gain a deeper understanding of the rising housing prices across the United States and to present a common trend that can help better explain why this might be true.

The data gathered here will show the trend specifically between median household income and house prices. In addition to that, the study will include housing price trends in certain regions.

Methods

Data Collection: The housing data has been collected from Zillow. The analysis has been performed on county, state, and metro data for the Single-Family Homes data from the Zillow Home Value Index (ZHVI).

The income data has been collected from HDPulse, using the state and county data for Median Household Income (2018-2022).

Data Cleaning: Housing rates datasets:

- Kept only one month's data for each year from 2000-2024 and renamed each column with only the year.
- Inserted another column and used VLOOKUP to give the full state name for each row because only the state abbreviation was given.

Data Enrichment: Housing rates datasets:

- Table format was used (also applicable to Income data)
- Inserted another column and used VLOOKUP to give the full state name for each row because only the state abbreviation was given.
- Added one to the size rank due to the rank starting at 0.

Data Exploration and Analysis using Excel:

- Sorting and filtering of data based on what graphs were needed.
- Maps displayed a visualization that was neat and easy to understand.
- Column, bar, and statistics charts were used to explore and create graphs for data.

Data Exploration and Analysis using Python Libraries:

Pandas

CSV data was imported into Python and data was sorted in ascending/descending order. The first few rows of data were viewed, and based on the structure, data was further sorted to perform the analysis. Certain indexes were modified in a DataFrame. Some specific pieces of text in the data set were replaced in a shorter form to accommodate data visualizations. The correlation coefficient was also found and was displayed on some of the visualizations. Some rows of data had missing values, so the row was deleted. Outliers were also deleted from the data because the visualizations would be skewed.

Matplotlib

Visualizations had the axes of the graph labeled and the graphs were titled. The graphs were also given a specific type of style, and text was added to the graph. The color, font size, and linewidth were also set. Line graphs, horizontal graphs, and bar graph visualizations were used to analyze data in a few charts. The figure size was also adjusted, and a grid was added to the plot, if it was not already there with the style. Additionally, the visualization was adjusted to fit in the specific area and prevented overlaps.

Seaborn

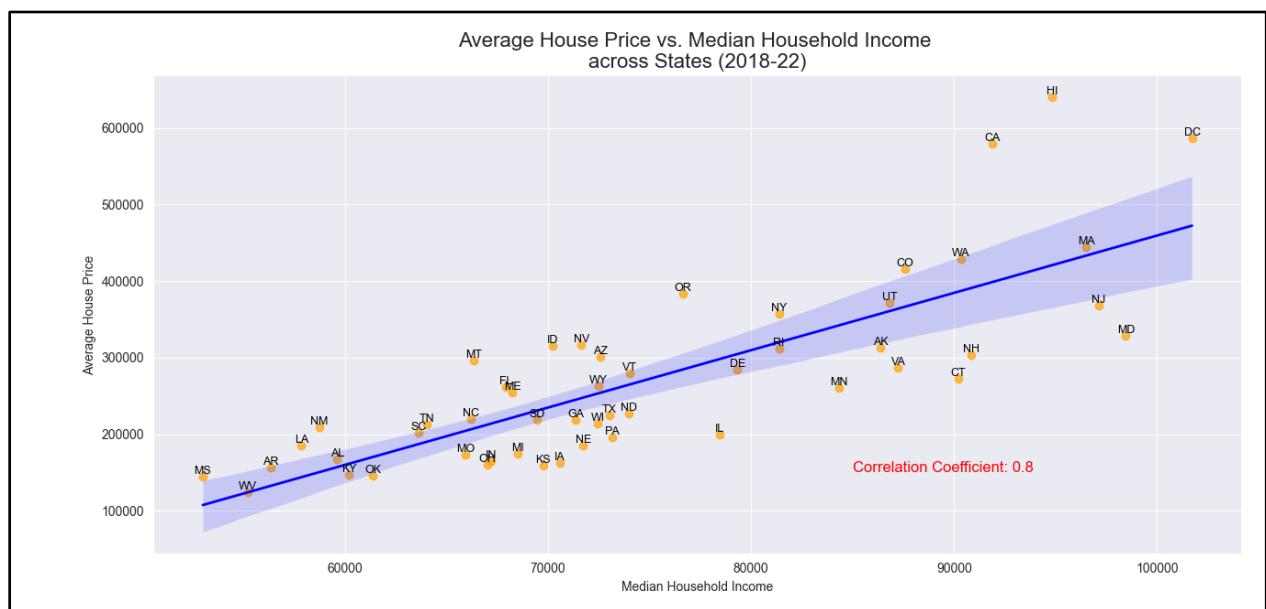
Seaborn was used for more concise and advanced visualizations. This included a scatterplot which was created with the regression line for correlational analysis. The heatmap was used to provide a visualization for multiple places over many years and was neatly displayed.

Results

Average House Price vs. Median Household Income across States (2018-22):

The following graph illustrates the correlation between average house price and median household income across states in the United States from 2018 to 2022. States such as California and Hawaii are significantly above the regression line, which could indicate that a lack of affordable housing is in those areas. In contrast, there are states below the regression line, such as Connecticut, due to confounding variables.

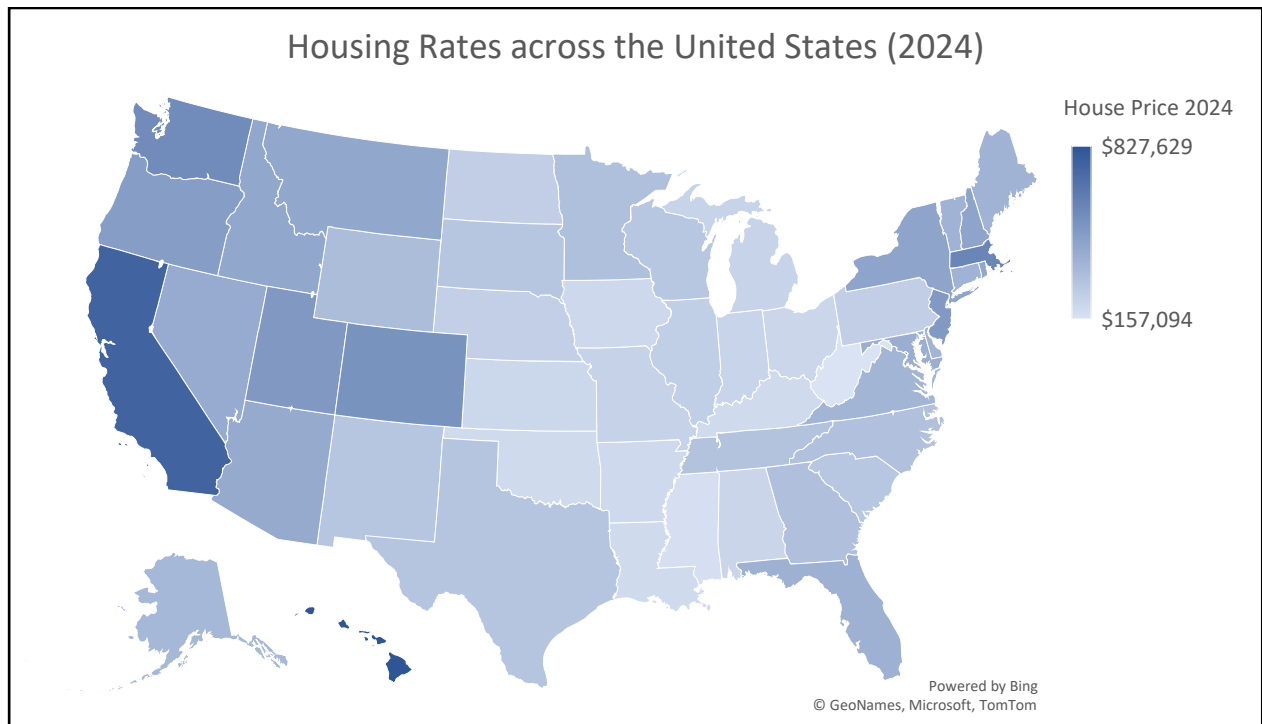
The correlation coefficient between house price and median household income across states is 0.8, a strong positive correlation which suggests that when income increases, house prices tend to increase. This does not necessarily mean that the change in income causes the change in house prices because there can be confounding variables, such as inflation in a certain region.



Correlation coefficients rounded to the nearest hundredth.

Housing Rates across the United States (2024):

The map below shows a visualization of the average single-home price across states in the United States for 2024. Darker areas indicate a higher house price, while lighter areas indicate a lower price. For example, California has high housing rates, which could be an issue especially considering California's population.

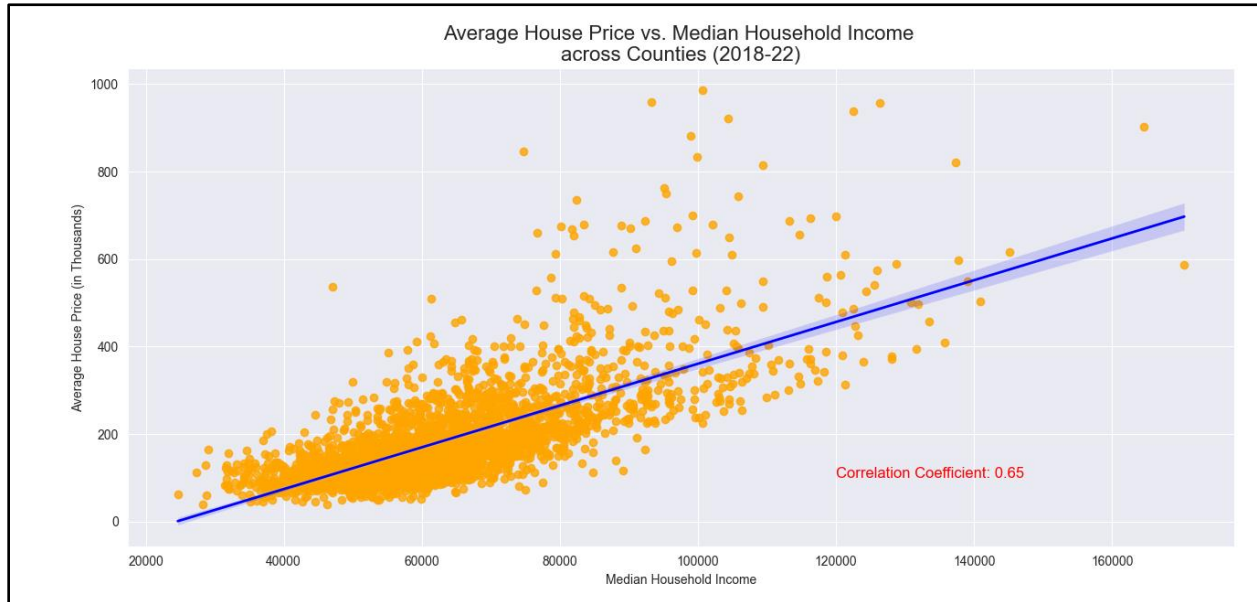


Housing rates rounded to the nearest hundredth.

Average House Price vs. Median Household Income across Counties (2018-22):

The graph below depicts the correlation between average house price and median household income across counties in the United States from 2018 to 2022.

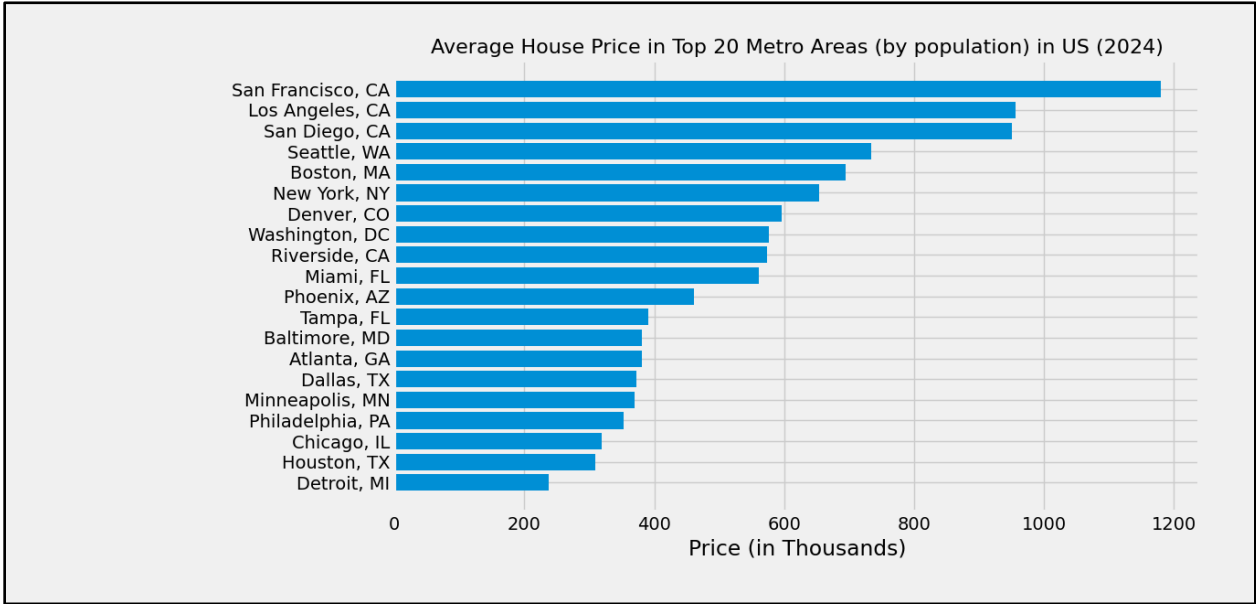
The correlation coefficient is 0.65, which shows that the correlation between average house price and median household income is a moderate positive correlation. Although most values seem to be close to the regression line (line of best fit), the correlation coefficient supports that there could be other factors which influence the affordability in these counties.



Correlation coefficients rounded to the nearest hundredth.

Average House Price in Top 20 Metro Areas (by population) in US (2024):

The following graph represents the average house price in the top twenty metropolitan areas (by population) in the United States for 2024.



Price rounded to the nearest hundredth.

The average house price in these regions were high, especially in San Francisco, Los Angeles, and San Diego because they all exceed \$900,000. These are areas that are the most and heavily populous in the country, which can affect many people. Based on the population data from 2023 U.S. Census Bureau data, the top twenty metro areas in 2023 made up 37.6% of the US population, which is a substantial portion of the population.

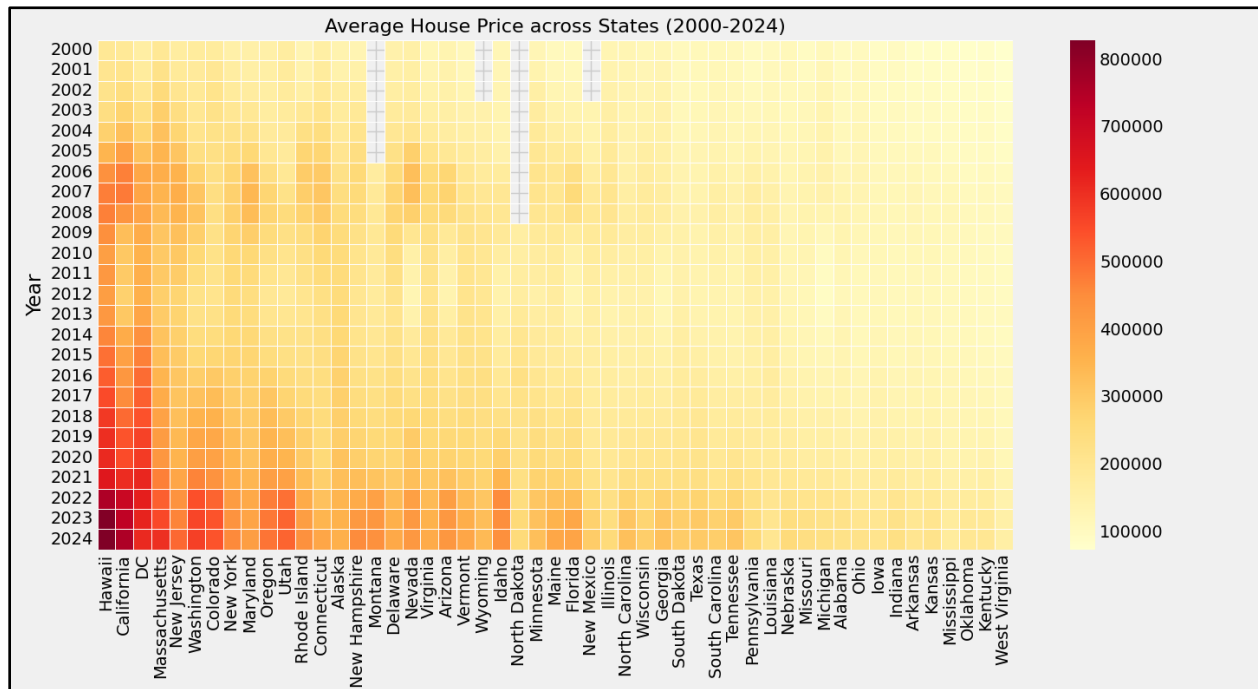
The following table shows the average house price in the top twenty metropolitan areas (by population) in the United States for 2024.

Metro Area	Price (2024)	Metro Area	Price (2024)
San Francisco, CA	\$1,180,382.02	Phoenix, AZ	\$460,629.02
Los Angeles, CA	\$955,529.93	Tampa, FL	\$390,076.28
San Diego, CA	\$950,385.83	Baltimore, MD	\$381,320.63
Seattle, WA	\$734,347.86	Atlanta, GA	\$380,403.97
Boston, MA	\$694,012.91	Dallas, TX	\$372,084.92
New York, NY	\$653,788.14	Minneapolis, MN	\$370,196.86
Denver, CO	\$595,809.36	Philadelphia, PA	\$352,806.45
Washington, DC	\$575,677.90	Chicago, IL	\$319,360.79
Riverside, CA	\$573,072.40	Houston, TX	\$309,289.25
Miami, FL	\$560,234.01	Detroit, MI	\$237,718.57

Price rounded to the nearest hundredth.

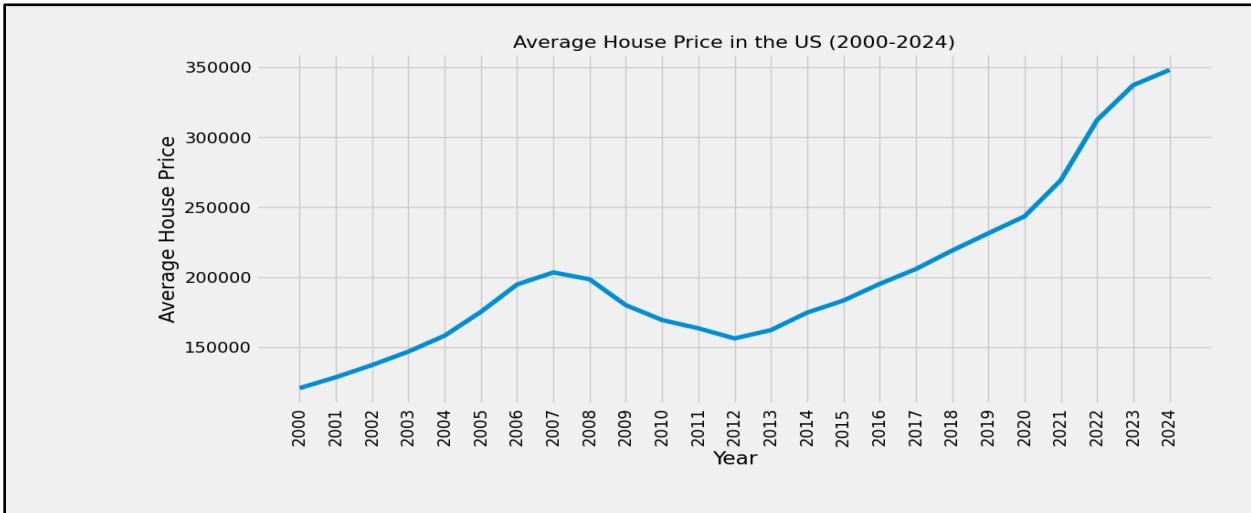
Average House Price across States (2000-2024):

The following visualization, a heatmap, shows the average house price across states throughout the years 2000 to 2024.



Average House Price in the US (2000-2024):

The graph below shows how the average house price in the United States changes from 2000-2024.



Price rounded to the nearest hundredth.

Just like in the heatmap, the house price typically increases, but there are exceptions to it, such as the drop in house prices from 2007 to 2012. Ever since the pandemic begun, the price has been increasing at a much more rapid rate, especially from 2020 to 2022, as supported in the above graph.

This next table shows the average house price in the United States from 2000-2024.

Year	Average House Price	Year	Average House Price
2000	\$120,663.19	2013	\$162,116.55
2001	\$128,444.26	2014	\$174,621.12
2002	\$137,155.91	2015	\$183,217.76
2003	\$146,752.44	2016	\$195,112.21
2004	\$158,141.48	2017	\$205,738.90
2005	\$175,146.35	2018	\$218,969.04
2006	\$194,761.83	2019	\$231,314.54
2007	\$203,342.53	2020	\$243,415.07
2008	\$198,427.72	2021	\$269,205.24
2009	\$179,930.44	2022	\$312,191.84
2010	\$169,325.83	2023	\$337,214.51
2011	\$163,405.96	2024	\$348,089.25
2012	\$156,172.69		

Price rounded to the nearest hundredth.

Conclusions

Housing prices across the United States have been rising over time and the disproportion between average house price and median household income has been increasing. In the housing market, supply does not meet demand. According to data, affordability is becoming increasingly difficult as time passes.

There is a strong positive correlation (0.8) between the average house price and median household income across states. There are a remarkable number of states that have extremely unaffordable prices compared to income, such as California and Hawaii. On the other hand, there is a moderate positive correlation (0.65) between the average house price and median household income across counties, which can show that other factors besides income contribute to affordability.

Compared to the rest of the country, housing rates are extremely high in the top metropolitan areas. That is significant because the top twenty metro areas consist of almost 40% of the nation's population. The trend of house prices increasing has continued throughout the years across all states. House prices typically increased nationwide over the past 25 years, especially at a rapid rate post pandemic.

In summary, the research done above shows that rising home prices and income being disproportionate has become a major affordability issue. This trend is particularly shown greater in metropolitan areas where house prices exceed median household incomes.

Next Steps

The next steps are for the government to act. Additional and more comprehensive data would be needed to predict the long-term effects of this crisis. Without intervention, the economy can get unstable, making inflation worse, which will lead to higher house prices. This can put millions of Americans in financial hardship. With proper action taken, the affordable housing rates can increase.

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